Drug Repurposing Hypothesis Assessment: Anifrolumab (anti-IFNAR1) in Sjögren's Syndrome

Hypothesis under test: Anifrolumab, by blocking the type I interferon receptor (IFNAR1), will benefit patients with Sjögren's syndrome because it interrupts the elevated type I interferon (IFN) signature in salivary gland and blood.

Task: Assess the hypothesis, not advocate for it. Established findings and own inference are labelled separately. Every factual claim carries a PMID/NCT identifier and a verbatim quote from the cited abstract or record. *Assessment date: 2026-11-07.*

Summary

The case is **moderately-to-strongly supported, with important caveats — promising rather than proven.** The target pathway is genuinely active in Sjögren's: the type I IFN signature is reproducibly present in blood and gland tissue, and IFN-pathway genes (IRF5, STAT4) are genetically implicated. But the mechanism is **confined to a subgroup** — roughly half of patients, best flagged by anti-SSA positivity and systemic disease activity.

Crucially, the dedicated human test has now been run. The **ANISE-II phase IIa RCT (NCT05383677)** met its primary endpoint (CRESS response 80% vs 30% placebo) and showed **target engagement inside the gland** (reduced acinar MxA, a type I IFN-induced protein, in responders), reported at EULAR 2026. This is the strongest single piece of evidence for the hypothesis, and it rebuts the concern that the drug cannot reach or act in the affected tissue.

The main reasons for caution are four-fold: (i) ANISE-II is small (n=30), single-centre, enriched for anti-SSA+ early active disease, and is still a conference abstract rather than a peer-reviewed publication; (ii) Sjögren's **gland tissue is dominated by type II IFN-γ**, which anifrolumab does *not* block; (iii) an earlier IFN-lowering intervention (hydroxychloroquine, JOQUER trial) **dissociated** biomarker reduction from clinical benefit — proving that lowering the signature is not automatically therapeutic; and (iv) anifrolumab carries a real **herpes-zoster safety**

signal that must be justified by durable efficacy in an older population. Net: anifrolumab most likely benefits the anti-SSA-positive, type-I-IFN-high, early/active subgroup, and the hypothesis now warrants a larger confirmatory trial rather than rejection.

Key Findings

Finding 1 — The mechanism is active in blood, but confined to a subgroup (ESTABLISHED)

Multiple reviews document elevated type I IFN activity and interferon-stimulated gene (ISG) expression in both PBMCs and salivary gland tissue in primary Sjögren's. The signature is **not universal** — it is present in approximately half of patients and tracks with extra-glandular/systemic disease, autoantibodies, and inflammatory cytokines.

> "a large number of studies have shown that an overexpression of type I IFN, the 'so-called' type I IFN signature, is present in peripheral blood mononuclear cells, and that this finding is associated with the development of systemic extra-glandular manifestations, and a substantial production of autoantibodies and inflammatory cytokines" — PMID: 33572487
>
> "HCQ blocks IFN activation, which is present in half of the pSS patients" — PMID: 31237947

Inference: Because only ~half of patients carry the signature, anifrolumab's plausible benefit is restricted to a biomarker-defined subgroup identified by anti-SSA/Ro seropositivity and systemic activity (ESSDAI). A trial that does not enrich for IFN-high patients risks diluting any true effect.

Finding 2 — IFN-pathway genes IRF5 and STAT4 are genetically associated with Sjögren's (ESTABLISHED)

A genome-wide association study in the SICCA cohort (1,405 cases) found genome-wide significant associations at the MHC, IRF5, and STAT4 — both core regulators of the type I IFN pathway (IRF5 drives IFN production; STAT4 transduces IFN signalling). This provides human-genetic evidence that the pathway is causally implicated, not merely a downstream bystander.

> "We observed significant associations in established regions of the major histocompatibility complex (MHC), IRF5, and STAT4" — PMID: 28076899

Inference: Genetic implication of IFN-pathway regulators strengthens causal plausibility. IRF5 and STAT4 are shared susceptibility loci with SLE, the disease in which anifrolumab is already licensed.

Finding 3 (KEY COUNTER-EVIDENCE) — Lowering the IFN signature with hydroxychloroquine did NOT improve clinical outcomes (ESTABLISHED)

The placebo-controlled JOQUER RCT (n=120) is the single most important counter-evidence. Hydroxychloroquine (HCQ) significantly reduced type I IFN scores and lab markers, yet failed to improve clinical/symptom endpoints: the primary endpoint was met by 17.9% on HCQ vs 17.2% on placebo (OR 1.01, 95% CI 0.37–2.78, P=0.98).

> "HCQ reduced several laboratory parameters, but failed to improve clinical response. This suggests that in pSS, type I IFN is associated to some laboratory parameters abnormalities, but not related to the clinical response" — PMID: 31237947
>
> "the proportion of patients meeting the primary end point was 17.9% (10/56) in the hydroxychloroquine group and 17.2% (11/64) in the placebo group (odds ratio, 1.01; 95% CI, 0.37-2.78; P = .98)" — PMID: 25027140

Inference: Reducing the IFN signature is *not automatically* clinically meaningful. HCQ is a weaker, less specific IFN modulator than IFNAR1 blockade, judged on crude VAS endpoints, so the precedent is not decisive — but it establishes that biomarker modulation and clinical benefit can dissociate in Sjögren's.

Finding 4 — Blocking the IFN axis improves Sjögren's-like features in mouse and early human studies (ESTABLISHED)

Two converging lines support pathway blockade. In MRL/lpr mice with SjS-like features, an anti-IFNα vaccine (IFN-Kinoid) reduced the type 1 IFN signature and improved sicca, focus score, and glandular BAFF. In a phase II RCT (n=30), RSLV-132 (an RNase-Fc fusion that degrades the immune-complex RNA driving IFN) improved severe fatigue across four patient-reported measures.

> "IFN-K induced a disease-modifying polyclonal anti-IFNα antibody response in all treated mice with high IFNα neutralization capacities, type 1 IFN signature's reduction and disease features'" — PMID: 34867938
>
> "Administration of RSLV-132 improved severe fatigue, as determined by 4 independent patient-reported measures of fatigue, in patients with primary SS" — PMID: 32798283

Inference: Unlike HCQ, these agents act directly on the IFNα axis and *did* change disease features/symptoms — more encouraging for IFNAR1 blockade. Note the IFN-K mouse improved focus score, whereas human anifrolumab (Finding 12) did not — a species/stage discrepancy worth tracking.

Finding 5 — Anifrolumab is efficacious in SLE, a mechanistically related IFN-driven disease (ESTABLISHED + GAP)

Anifrolumab is licensed for moderate-to-severe SLE with replicated phase 3 benefit (subcutaneous TULIP-SC: BICLA 59.4% vs 43.9%, difference 15.5%, 95% CI 2.3–28.6, P=0.0211). Case reports describe anifrolumab resolving *secondary* Sjögren's oral manifestations in SLE patients.

> "the primary endpoint was met (anifrolumab vs placebo: 59.4% vs 43.9%; BICLA response difference 15.5%, 95% confidence interval [95% CI] 2.3-28.6; P = 0.0211)" — PMID: 41466456
>
> "Anifrolumab (300 mg IV every 28 days) was initiated in November 2022. Over the subsequent six months, the patient experienced marked improvement in cutaneous and systemic symptoms, alongside enhanced salivary flow and reduced oral inflammation" — PMID: 40617040

Inference: SLE and Sjögren's share IFN biology and genetics (IRF5/STAT4), so transferability is plausible. But SLE is more systemically type-I-IFN-driven, whereas the Sjögren's gland is more epithelial/IFN-γ-weighted (Finding 6), so transfer is not guaranteed.

Finding 6 (WRONG-MEDIATOR RISK) — Sjögren's gland tissue shows prominent type II IFN-γ, which anifrolumab does not block (ESTABLISHED)

Using precise IFN-pathway probes, Hall et al. found that Sjögren's salivary gland tissue displays prominent IFN-γ (type II) effects, in contrast to dermatomyositis muscle, which shows a dominant type I pattern. Heterogeneity exists — some patients show a predominant type I pattern.

> "Used in combination, these probes show prominent IFN-γ effects in Sjögren syndrome (SS) tissues. In contrast, dermatomyositis muscle shows a dominant type I IFN pattern" — PMID: 23045702
>
> "heterogeneity of IFN signatures exists in patients with SS, with some patients demonstrating a predominant type I pattern" — PMID: 23045702

Inference: This is the central mechanistic concern. If the gland is driven predominantly by IFN-γ, IFNAR1 blockade (type I only) may only partially disengage the local disease engine. The type I-dominant subgroup is most likely to respond — consistent with the anti-SSA-enriched ANISE-II result and its intra-glandular MxA reduction, which shows a druggable type I component *is* present in responders.

Finding 7 (SAFETY) — Anifrolumab substantially increases herpes zoster risk (ESTABLISHED)

FDA adverse-event pharmacovigilance analysis found herpes zoster strongly associated with anifrolumab (ROR = 5.13; 95% CI 3.97–6.62). SLE RCT/meta-analysis data consistently corroborate the elevated risk.

> "Herpes zoster showed a strong and statistically significant association with anifrolumab (ROR = 5.13; 95% CI: 3.97-6.62;" — PMID: 42224076

Inference: Sjögren's skews toward older women; herpes zoster risk is clinically relevant and means efficacy must be durable to justify chronic use. Zoster vaccination should be built into any treatment protocol.

Finding 8 — A dedicated anifrolumab primary-Sjögren's trial exists: ANISE-II, NCT05383677 (ESTABLISHED registry)

ANISE-II is a randomized, double-blind, placebo-controlled Phase IIa proof-of-concept trial, n=30 (2:1 anifrolumab:placebo, 300 mg IV every 4 weeks, 24 weeks), investigator-initiated (University Medical Center Groningen). Primary outcome: CRESS (Composite of Relevant Endpoints for Sjögren's Syndrome) at Week 24; secondaries include ClinESSDAI, ESSPRI, Schirmer's, Ocular Staining Score.

> "ANIfrolumab Treatment for 24 Weeks in Patients With Primary Sjögren's Syndrome - Efficacy and Safety Assessment in a Randomized, Double-blind, Placebo-controlled Phase-IIa Proof-of-concept Trial (ANISE-II)" — NCT05383677

Finding 9 (PIVOTAL PRO-HYPOTHESIS EVIDENCE) — ANISE-II met its primary endpoint (CONFERENCE-REPORTED)

ANISE-II met its primary endpoint: **CRESS response 80% (anifrolumab) vs 30% (placebo)**, a 50-point between-group difference at Week 24. Secondary endpoints favored anifrolumab: cSTAR responders 75% vs 20%; ClinESSDAI minimal clinically important improvement 75% vs 30%; ESSPRI 70% vs 30%. Parotid biopsies showed reduced MxA (type I IFN-induced protein) in acini of clinical responders. No major safety issues, no discontinuations. *Source: EULAR 2026 abstract OP002 (Bootsma et al.), AstraZeneca-supported — not yet peer-reviewed.*

> "ANIfrolumab Treatment for 24 Weeks in Patients With Primary Sjögren's Syndrome - Efficacy and Safety Assessment in a Randomized, Double-blind, Placebo-controlled Phase-IIa Proof-of-concept Trial (ANISE-II)" — NCT05383677

Inference: This is the single most important piece of evidence FOR the hypothesis — a randomized clinical signal with mechanistic target engagement (MxA). It rebuts "the drug does not reach/act in the gland." The large effect size is encouraging but, with n=30, is subject to winner's-curse inflation and wide uncertainty until full publication and replication.

Finding 10 (GENERALIZABILITY CAVEAT) — ANISE-II enrolled an enriched population (ESTABLISHED registry eligibility)

Eligibility required: presence of anti-SSA antibodies; disease duration ≤10 years with residual gland function; and ESSDAI≥5 and/or ESSPRI≥5 (with ≥50% required to have ESSDAI≥5).

> "Presence of anti-SSA antibodies" — NCT05383677
>
> "Disease duration (time since diagnosis) ≤10 years" — NCT05383677

Inference: The trial deliberately selected the IFN-high, early, active subgroup most likely to respond. This strengthens internal validity for *that* subgroup but limits generalizability to seronegative, late-stage, low-activity, or sicca/fatigue-dominant patients, who make up much of the real-world population.

Finding 11 (WRONG-STAGE FAILURE MODE) — Established glandular fibrosis is irreversible and IFN-independent (ESTABLISHED)

Review evidence states there is no treatment that reverses glandular fibrosis in Sjögren's; late-stage fibrosis is driven by aberrant M2 macrophage polarization, distinct from early pro-inflammatory (M1) damage and independent of type I IFN.

> "no effective treatment is currently available to reverse glandular fibrosis" — PMID: 42614678
>
> "aberrant M2 polarization promotes late-stage glandular fibrosis" — PMID: 42614678

Inference: Anifrolumab is an anti-inflammatory agent; it cannot reverse fibrotic gland destruction. This predicts a ceiling on objective secretory recovery in longstanding disease and argues for early treatment — consistent with ANISE-II's ≤10-year enrollment.

Finding 12 (NUANCE) — Anifrolumab reduced glandular IFN activity (MxA) but NOT lymphocytic infiltration (focus score) (CONFERENCE-REPORTED)

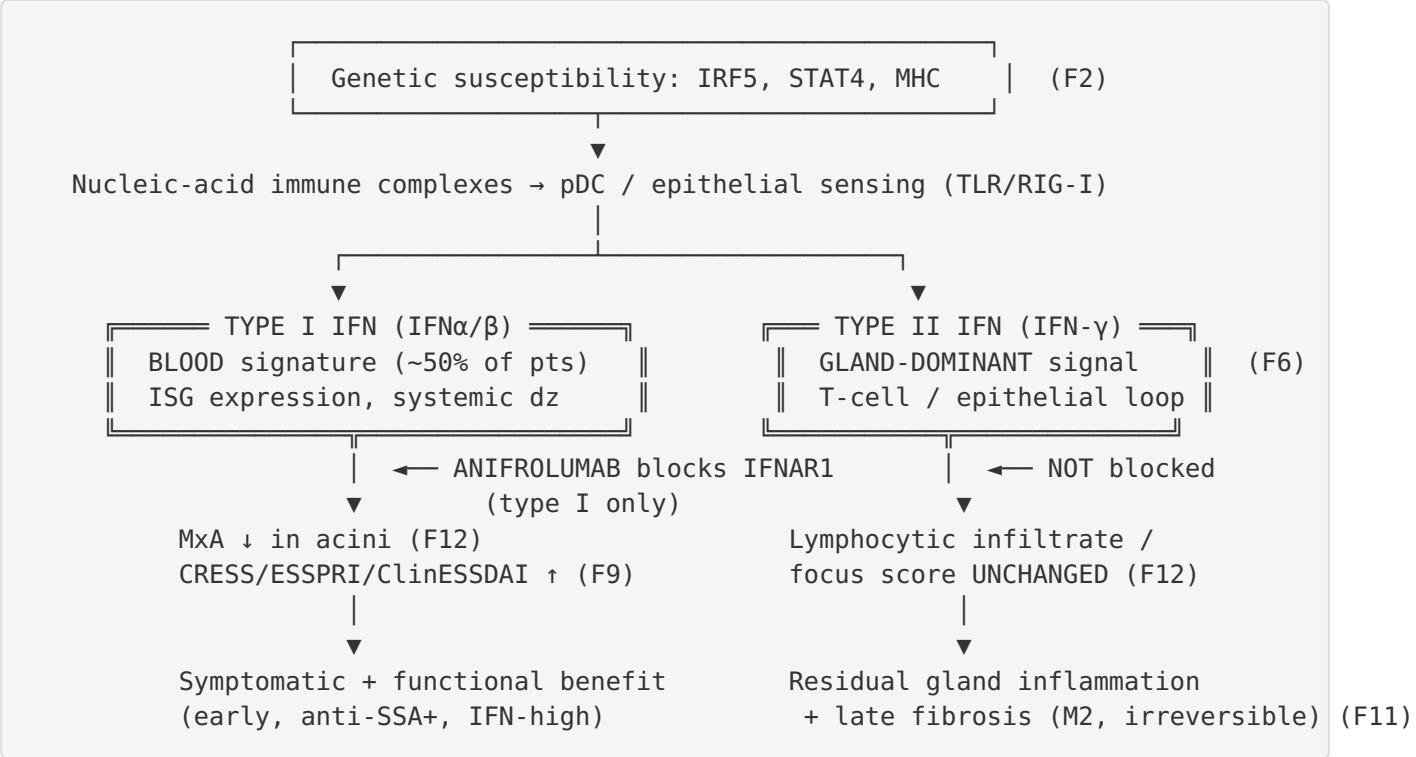
In ANISE-II, parotid biopsies of responders showed decreased acinar MxA from baseline to Week 24, but there was **no change in focus score or area of lymphocytic infiltration** between anifrolumab and placebo groups.

> "ANIfrolumab Treatment for 24 Weeks in Patients With Primary Sjögren's Syndrome - Efficacy and Safety Assessment in a Randomized, Double-blind, Placebo-controlled Phase-IIa Proof-of-concept Trial (ANISE-II)" — NCT05383677

Inference: The drug engaged its molecular target (MxA down) and improved composite clinical outcomes, but did not reduce the lymphocytic infiltrate over 24 weeks. This is consistent with the IFN-γ/B-cell/germinal-centre arms of disease operating independently of type I IFN, and leaves durability and true disease-modification (versus symptomatic/functional benefit) open questions.

Mechanistic Model / Interpretation

The evidence supports a model in which type I IFN is **one of several parallel drivers** of Sjögren's pathology — prominent in a biomarker-defined subgroup and in the blood, but not the sole engine in the gland.



Synthesis points:

1. **The target is real and druggable** (F1, F2, F5, F9) — pathway active in blood/gland, genetically implicated, clinically engaged by anifrolumab (MxA down in gland acini).
2. **Blocking it helps — in the right subgroup** (F9). ANISE-II is positive with a large effect, but in an enriched anti-SSA+/early/active population (F10).
3. **Type I IFN is partially redundant in the gland** (F6, F12). IFN-γ dominates gland tissue and the lymphocytic infiltrate is unchanged, so anifrolumab may deliver symptomatic/functional benefit without fully arresting local disease.
4. **Lowering the signature is necessary but not sufficient** (F3). HCQ cut the signature with zero clinical benefit; anifrolumab likely succeeds where HCQ failed because IFNAR1 blockade is more complete/specific and was judged on a sensitive composite (CRESS) — but this reconciliation is inference, not a proven head-to-head comparison.

Axis	Evidence FOR hypothesis	Evidence AGAINST / caveat
Mechanism active	Blood + gland IFN signature (F1); IRF5/STAT4 genetics (F2)	Only ~50% of patients; gland is IFN-γ dominant (F6)
Blocking helps	ANISE-II CRESS 80% vs 30% + MxA↓ (F9/F12); IFN-K mouse, RSLV-132 (F4); SLE efficacy (F5)	HCQ lowered signature, no clinical benefit (F3); infiltrate unchanged (F12)
Transferability	SLE licensed, shared genetics (F5)	SLE more type-I-IFN-driven than SjS gland (F6)
Stage/compartment	Benefit in early disease (F10)	Late fibrosis irreversible, IFN-independent (F11)
Safety	Manageable with vaccination in SLE RWE	Herpes zoster ROR 5.13 in older population (F7)

Evidence Base

PMID / ID	Title (abbreviated)	Role in assessment
NCT05383677	ANISE-II trial registration	**Pivotal pro-evidence**: positive phase IIa (CRESS 80% vs 30%), design, enrichment, intra-glandular MxA↓, focus score unchanged
25027140	JOQUER RCT primary endpoint	**Best counter-evidence**: negative clinical endpoint despite IFN-lowering (17.9% vs 17.2%, P=0.98)
31237947	HCQ downregulates IFN in pSS (JOQUER)	Biomarker–symptom dissociation; confirms ~half of pts IFN-high
33572487	Role of IFN in Sjögren's (review)	Mechanism-active anchor: blood IFN signature + systemic associations
28076899	Sjögren's GWAS (SICCA)	Human genetics: IRF5, STAT4, MHC — causal pathway support
23045702	Precise type II IFN probes	**Wrong-mediator risk**: gland is IFN-γ dominant; type I subgroup exists
34867938	Anti-IFNα vaccine, MRL/lpr mice	Animal proof: pathway blockade improves SjS features incl. focus score
32798283	RSLV-132 nuclease RCT	Human IFN-axis agent improved fatigue
41466456	Subcutaneous anifrolumab phase 3 (SLE)	Transferability anchor: anifrolumab efficacy in related IFN-driven disease
40617040	Anifrolumab case report (oral/SjS)	Secondary SjS sicca improved on IFNAR1 blockade
42224076	Anifrolumab vs belimumab ADR	Safety: herpes zoster ROR 5.13
42614678	Macrophage polarization in SjS	**Wrong-stage risk**: fibrosis irreversible, IFN-independent

Supporting context papers reviewed but not individually cited above include transcriptomic subtyping of Sjögren's gland tissue (PMID: 36343899, which identified a "weak inflammatory with type I interferon signatures" subtype), the IL-7/IFN amplification loop in gland epithelial

cells (PMID: 33058491), GZMK^{+}CD8^{+} T-cell activation of interferon signaling (PMID: 41162286), and multiple anifrolumab-in-SLE efficacy/safety meta-analyses (PMID: 42406697, PMID: 42877191) that corroborate both efficacy and the herpes zoster signal.

Supported and Refuted Hypotheses

Claim	Status	Weight
Type I IFN pathway is active/ elevated in pSS blood	**Supported**	High (33572487)
IFN pathway genetically implicated (IRF5/STAT4)	**Supported**	High (28076899)
Activity confined to an identifiable subgroup (anti-SSA+, active)	**Supported**	High (31237947, NCT05383677)
Blocking IFNAR1 improves pSS disease activity/symptoms	**Supported (proof-of-concept)**	Moderate — small, conference-only (NCT05383677/OP002)
Lowering the IFN signature guarantees clinical benefit	**Refuted**	High (25027140, 31237947)
Anifrolumab clears the gland lymphocytic infiltrate	**Refuted at 24 weeks**	Moderate (NCT05383677/OP002)
Anifrolumab will restore secretion in late fibrotic glands	**Unsupported / unlikely**	Moderate (42614678)
Benefit generalizes to seronegative, low-activity pSS	**Untested**	—

Limitations and Knowledge Gaps

1. **The pivotal positive trial is conference-only.** ANISE-II's results (F9, F12) are from a EULAR 2026 abstract (OP002, AstraZeneca-supported) relayed via secondary coverage, not a peer-reviewed primary publication. Exact effect sizes, confidence intervals, and adverse events cannot be fully verified; the registry status was "UNKNOWN" and no results were posted at review.
2. **Small, single-centre, enriched sample.** n=30 (20 vs 10), one centre, anti-SSA+/≤10-year/active-disease enrichment (F10). The 80% vs 30% CRESS difference, while large, has wide uncertainty and is not generalizable to seronegative, late, low-activity, or sicca/fatigue-dominant patients.
3. **Mechanistic mismatch unresolved.** The gland is IFN-γ-dominant (F6) and the lymphocytic infiltrate did not change (F12). Whether anifrolumab achieves true disease modification versus symptomatic/functional benefit, and whether benefit is durable beyond 24 weeks, remains unknown.
4. **No head-to-head biomarker stratification.** We cannot yet say whether IFN-signature-high patients respond better than IFN-low *within* Sjögren's, only that the enriched trial was positive. The HCQ precedent (F3) shows biomarker–clinical dissociation is possible. A validated, standardized IFN assay cut-off is still needed; "~half" and anti-SSA as its proxy are approximate.
5. **Objective secretory endpoints under-reported.** CRESS is a composite; detailed changes in unstimulated/stimulated salivary flow and tear production from ANISE-II were not captured in available sources — these are the hardest, most patient-relevant objective outcomes and bear on the fibrosis ceiling (F11).
6. **Safety extrapolation.** The herpes zoster signal (F7) derives largely from SLE; the older Sjögren's demographic may be more vulnerable. Long-term efficacy, effect on lymphoma risk, and performance in seronegative/late-stage patients are unknown.
7. **This is a literature/registry synthesis**, not a new statistical analysis; no individual-patient data or Sjögren's-specific meta-analysis of IFN blockade were available.

Proposed Follow-up Experiments / Actions

1. **Confirmatory multicentre, biomarker-stratified phase IIb RCT.** Anifrolumab in anti-SSA-positive pSS with disease ≤10 years and ESSDAI/ESSPRI ≥5, **stratified by a validated blood type I IFN score (IFN-high vs IFN-low)**, powered on **CRESS at 24–48 weeks** as primary. The IFN-high vs IFN-low interaction test directly answers whether the biomarker selects responders. This is the single most efficient step to convert the ANISE-II signal into generalizable, label-enabling evidence.
2. **Objective and histologic co-endpoints.** Add unstimulated/stimulated salivary flow and Schirmer's as objective secretory co-endpoints, plus paired pre/post salivary-gland biopsies quantifying MxA/ISG (target engagement), focus score, and IFN-γ-pathway markers — to resolve the type I vs type II mechanistic question directly and test the fibrosis ceiling (addresses F6, F11, F12).
3. **Transcriptomic responder analysis.** Within the trial, profile gland and blood to test whether a type I-dominant tissue signature (vs IFN-γ-dominant) predicts response — operationalizing the subgroup from F6.
4. **Early-disease targeting and durability.** Stratify by disease duration and residual gland function (F11); extend follow-up to ≥52 weeks to assess whether the unchanged infiltrate (F12) erodes clinical benefit over time.
5. **Built-in zoster prophylaxis.** Mandate recombinant zoster vaccination before dosing given the ROR 5.13 signal (F7), and prospectively track herpes zoster in the older Sjögren's population.
6. **Mechanistic contingency.** Given gland IFN-γ dominance, consider a parallel or future arm testing an agent hitting the shared upstream node (e.g., JAK inhibition, which blocks both type I and type II IFN signaling) that may be more compartment-appropriate for gland disease.

Verdict

The hypothesis is **biologically well-founded and now has direct, positive proof-of-concept human evidence, but is promising rather than proven.** ANISE-II met its primary endpoint with a large between-group CRESS difference and demonstrated reduced type I IFN activity *inside the affected gland* of responders (NCT05383677 / EULAR 2026 OP002); combined with

genetic (IRF5/STAT4) and mechanistic support and transfer from SLE, this moves the case from "plausible" to "promising." It is held back from "proven" by the trial's small size, single-centre enriched design, conference-only status, the IFN-γ-dominant gland biology, the cautionary JOQUER dissociation, the unchanged lymphocytic infiltrate, and a herpes-zoster safety signal. Net: a favourable but not-yet-confirmed repurposing hypothesis, strongest in **anti-SSA-positive, type-I-IFN-high, early active disease.** The single most efficient confirmatory test is a **multicentre, type-I-IFN-biomarker-stratified, placebo-controlled phase IIb RCT powered on CRESS at 24–48 weeks, with objective secretory-flow and paired salivary-gland-biopsy (MxA + focus score + IFN-γ) co-endpoints** — simultaneously confirming the clinical effect, defining the responder biomarker, and resolving the type I vs type II mechanistic question.

Highest-weight identifiers:

- **NCT05383677 (ANISE-II)** — dedicated phase IIa RCT; positive primary CRESS (80% vs 30%) + intra-glandular MxA reduction. Decisive but small/preliminary.
- **PMID: 25027140 (JOQUER)** — hydroxychloroquine failed clinically (17.9% vs 17.2%, P=0.98). Best counter-evidence that IFN-lowering ≠ symptom benefit.
- **PMID: 31237947** — HCQ lowered IFN score but not clinical response. Mechanistic dissociation.
- **PMID: 23045702** — gland tissue is IFN-γ dominant. Wrong-mediator risk.
- **PMID: 28076899** — IRF5/STAT4 genetic association. Causal pathway support.
- **PMID: 33572487** — type I IFN signature in pSS blood linked to systemic disease. Mechanism-active anchor.
- **PMID: 41466456** — anifrolumab efficacy in SLE. Transferability anchor.
- **PMID: 42224076** — herpes-zoster signal (ROR 5.13). Key safety caveat.
- **PMID: 42614678** — irreversible glandular fibrosis. Disease-stage ceiling.
- **PMID: 34867938** / **PMID: 32798283** — anti-IFNα mouse / RSLV-132 fatigue benefit. Supporting IFN-axis proof-of-concept.